



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

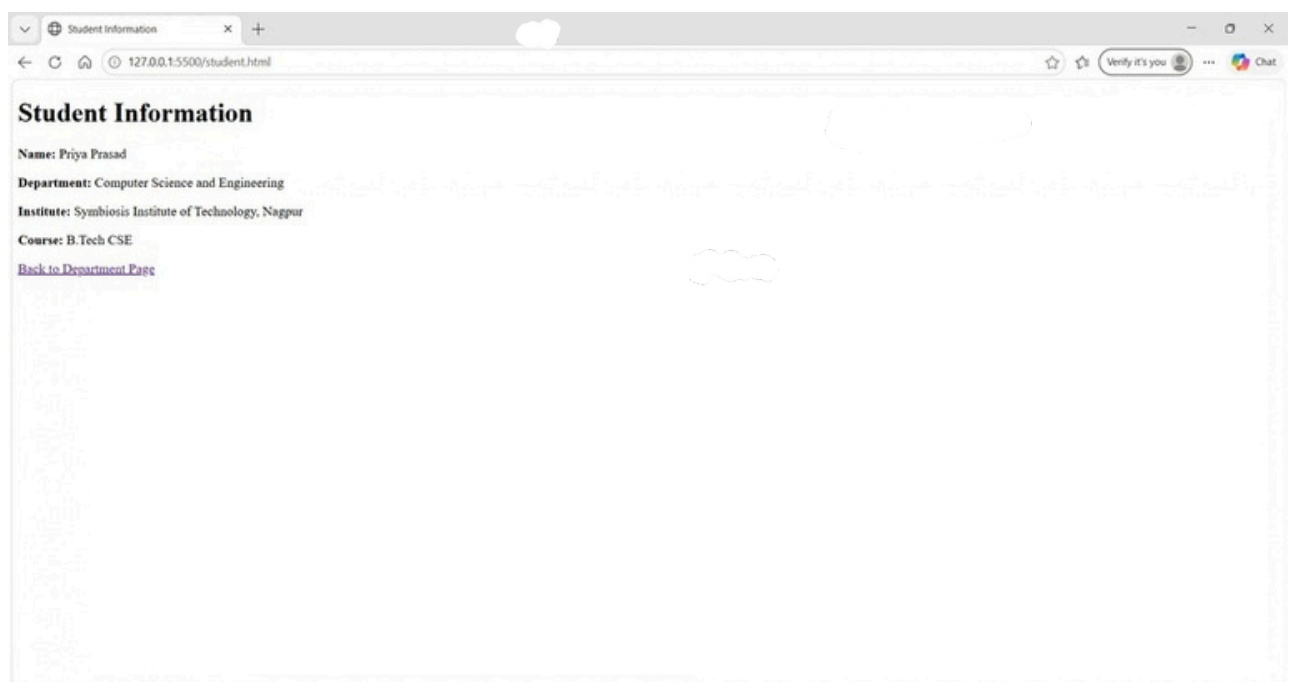
Case Study : 1

1. Title: Create a Webpage for College Department and Link the Student Information Page Using JavaScript Alert.

2. Software/Tools Required: HTML, VS Code.

3. Theory: This case study focuses on creating a college department webpage using HTML and JavaScript. The webpage displays information about the Computer Science and Engineering Department of SIT Nagpur. A separate student information page is linked to the department webpage using an HTML hyperlink. JavaScript is used to display a welcome message, "Welcome to SIT nagpur", in an alert box when the user clicks the Welcome button

4. Output:





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Conclusion : In this case study, we successfully created a college department webpage and linked it to a student information page. We also used JavaScript to display a welcome message in an alert box. This activity helped us understand HTML page linking and basic JavaScript alert functions.



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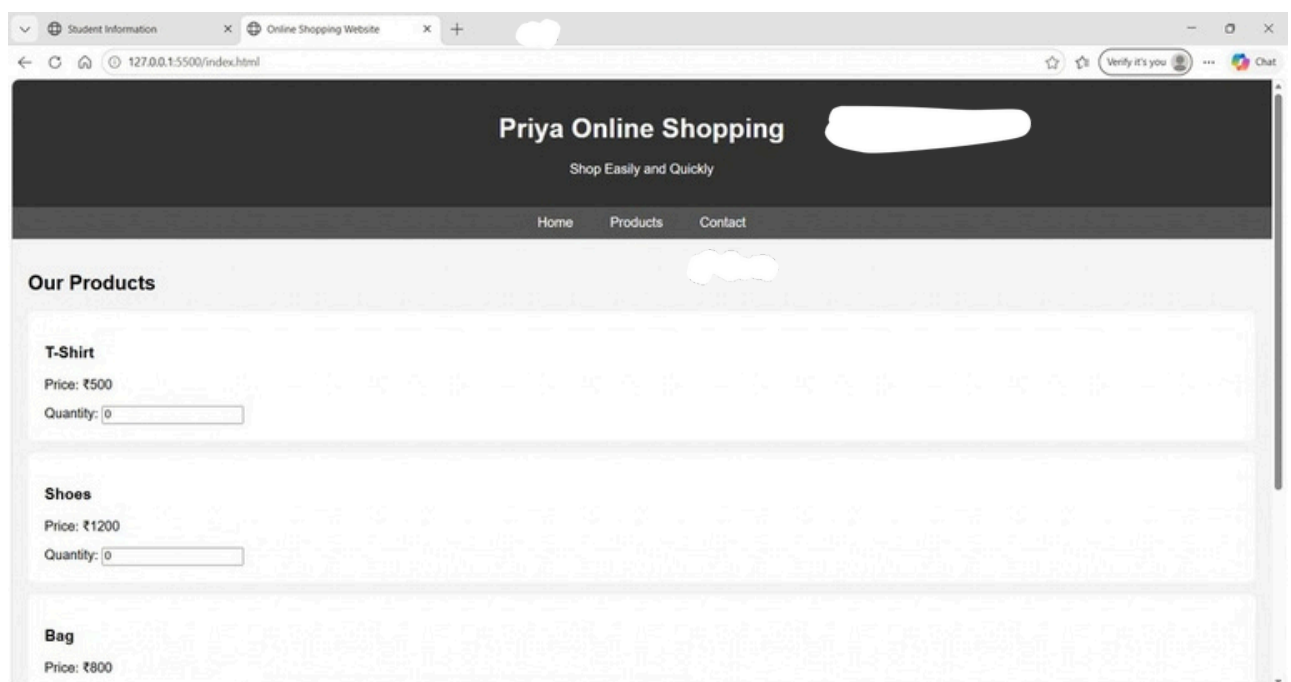
Case Study : 2

1. Title: Develop an Online Shopping Website Using HTML, CSS, HTML5 Tags, and JavaScript to Calculate the Bill.

2. Software/Tools Required: HTML, VS Code.

3. Theory: This case study focuses on developing an online shopping website using HTML, CSS, and JavaScript. HTML5 semantic tags such as `<header>`, `<nav>`, `<main>`, `<section>`, `<article>`, and `<footer>` are used to structure the webpage. CSS is used to design and style the website. JavaScript is used to accept product quantities, calculate the total price, add GST, and display the final bill dynamically on the webpage.

4. Output:





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Conclusion : In this case study, we successfully developed an online shopping webpage using HTML, CSS, and JavaScript. We learned how to use HTML5 semantic tags, apply CSS styling, and calculate the total bill using JavaScript. The webpage displays product details and generates the final bill based on the selected quantities.



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Case Study : 3

1. Title: Scholarship Eligibility Checker Using JavaScript

2. Software/Tools Required: HTML, VS Code.

3. Theory: The Scholarship Eligibility Checker is a web-based application developed using HTML, CSS, and JavaScript. It takes the student's marks and annual family income as input. JavaScript uses conditional statements (if-else) to check whether the student meets the given eligibility criteria. The result is displayed dynamically on the webpage using DOM manipulation.

4. Output:



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Conclusion : In this case study, we successfully developed a Scholarship Eligibility Checker using JavaScript. The application checks the student's marks and family income using conditional statements and displays the eligibility result. This activity helped us understand JavaScript conditions, user input, and DOM manipulation.



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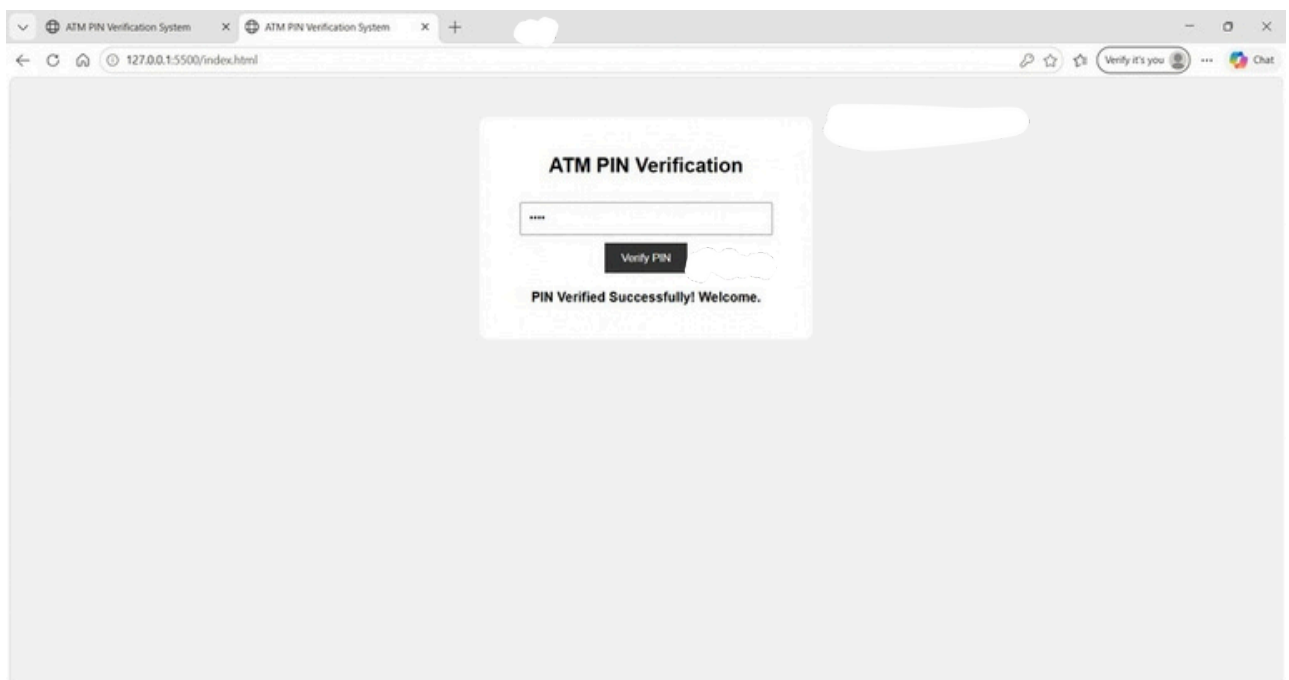
Case Study : 4

1. Title: ATM Card PIN Verification System Using JavaScript Functions, Closures, and Exception Handling

2. Software/Tools Required: HTML, VS Code.

3. Theory: The ATM Card PIN Verification System is developed using HTML, CSS, and JavaScript. It verifies the PIN entered by the user and checks whether it is correct. The program demonstrates different JavaScript concepts, including function declaration, function expression, arrow functions, variable scope, and closures. Exception handling using try-catch is used to handle invalid PIN input and maximum login attempts. The system displays appropriate messages based on the verification result.

4. Output:





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Conclusion : In this case study, we successfully implemented an ATM Card PIN Verification System using JavaScript. We learned how to use different types of functions, variable scope, closures, and exception handling. The program validates the entered PIN and displays suitable messages for successful and unsuccessful verification. This activity helped us understand the practical use of JavaScript functions and error handling.



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Case Study : 5

1. Title: Study of Arrays and Loops in JavaScript with Maximum and Minimum Value Calculation.

2. Software/Tools Required: HTML, VS Code.

3. Theory: Arrays in JavaScript are used to store multiple values in a single variable. Array methods such as `push()`, `pop()`, `shift()`, `unshift()`, `splice()`, and `slice()` are used to add, remove, and extract elements from an array. Methods like `map()`, `filter()`, `reduce()`, and `forEach()` help in processing and manipulating array elements. In this case study, an array of numbers is created, and JavaScript is used to find the maximum and minimum values using `Math.max()` and `Math.min()`.

4. Output:

```
Arrays and Loops in JavaScript
Original Array: 10,20,30,40,50
After push(60): 10,20,30,40,50,60
After pop(): 10,20,30,40,50
After unshift(5): 5,10,20,30,40,50
After shift(): 10,20,30,40,50
After splice(): 10,20,35,40,50
After slice(1,4): 20,35,40

forEach(): 10 20 35 40 50
map() - Doubled Array: 20,40,70,80,100
filter() - Numbers > 20: 35,40,50
reduce() - Sum: 155

Case Study
Array: 10,20,35,40,50
Maximum Value: 50
Minimum Value: 10
```



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Conclusion : In this experiment, we successfully studied arrays and loops in JavaScript. We implemented different array methods to add, remove, and manipulate elements. We also used `map()`, `filter()`, `reduce()`, and `forEach()` for array processing. The case study helped us find the maximum and minimum values from an array of numbers and understand the practical use of JavaScript array methods.



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Case Study : 6

1. Title: To Perform Basic String Operations, Reverse a String, and Count the Number of Vowels in a Paragraph Using JavaScript.

2. Software/Tools Required: HTML, VS Code.

3. Theory: String operations in JavaScript are used to manipulate and process text. In this experiment, JavaScript string methods such as `length`, `toUpperCase()`, `toLowerCase()`, `split()`, and `reverse()` are used to perform basic string operations. A for loop and the `includes()` method are used to count the number of vowels in a paragraph. HTML is used to create the webpage, CSS is used for styling, and JavaScript is used to perform operations and display the results dynamically.

4. Output:

String Operations in JavaScript

Enter a Paragraph:

I am in Symbiosis
I am in third year
I am a student
My name is Priya Prasad

Perform Operations

Results

Original String:
I am in Symbiosis I am in third year I am a student My name is Priya Prasad

String Length: 75

Uppercase:
I AM IN SYMBIOSIS I AM IN THIRD YEAR I AM A STUDENT MY NAME IS PRIYA PRASAD

Lowercase:
i am in symbiosis i am in third year i am a student my name is priya prasad



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Conclusion : In this experiment, we successfully performed basic string operations using HTML, CSS, and JavaScript. We learned how to reverse a string and count the number of vowels in a paragraph using loops and string methods. This experiment helped us understand string manipulation and dynamic result display in JavaScript.



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Case Study : 7

1. Title: To Demonstrate Event Handling in JavaScript Using Form Elements and Input Validation.

2. Software/Tools Required: HTML, VS Code.

3. Theory: Event handling in JavaScript is used to respond to user actions on a webpage. In this experiment, HTML is used to create a student registration form, CSS is used for styling, and JavaScript is used to access and manipulate form elements. Different events such as focus, change, and submit are implemented. JavaScript validates user input, checks for empty fields, and verifies the email format before displaying the result.

4. Output:

The screenshot shows a web browser window with multiple tabs. The active tab is titled 'JavaScript Event Handling'. The address bar shows the URL '127.0.0.1:5500/index.html'. The main content area displays a 'Student Registration Form' with the following fields:

- Name:** A text input field containing 'Priya prasad'.
- Email:** A text input field containing 'priyaprasad@gmail.com'.
- Course:** A dropdown menu with 'Computer Science' selected.
- Submit:** A green button labeled 'Submit'.

Below the form, a green message states 'Form submitted successfully!'.



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Conclusion : In this experiment, we successfully implemented event handling in JavaScript using a student registration form. We learned how to access form elements, handle focus, change, and submit events, and validate user inputs. This experiment helped us understand how JavaScript makes webpages interactive and user-friendly.



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Case Study : 8

1. Title: Online Job Application Form Using HTML, CSS, and JavaScript

2. Software/Tools Required: HTML, VS Code.

3. Theory: The Online Job Application Form is developed using HTML, CSS, and JavaScript. HTML is used to create the form structure and input fields such as name, email, phone number, qualification, and job role. CSS is used to style the form and improve its appearance. JavaScript is used to validate user inputs, check empty fields, validate email and phone numbers, and handle the form submission event. DOM manipulation is used to display success and error messages dynamically.

4. Output:

Online Job Application Form

Full Name: Priya Prasad

Email: priya.prasad.batch2024@sitnagpur.siu.edu.in

Phone Number: 9876234567

Qualification: B.Tech

Job Role: Software Developer

Apply Now

Application submitted successfully!



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Conclusion : In this case study, we successfully created an Online Job Application Form using HTML, CSS, and JavaScript. We implemented form validation, event handling, and dynamic message display. This activity helped us understand how JavaScript can be used to access form elements, validate user input, and improve the functionality of web forms.